

Pokhara University
Faculty of Science and Technology

Course Code.: CMP 557 (3 Credits)
Course Title: Internet Technology (3-0-0)
Nature of the course: Theory
Level: Master

Full marks: 100
Pass marks: 60
Total Lectures: 45 hrs
Program: MSc CS

1. Course Description

The Internet Technology syllabus for Graduate students is designed to provide an in-depth understanding of the core principles, architecture, protocols, and evolving technologies that power the modern Internet. This course explores the foundations of internet communication, including IP addressing, routing, DNS, and web technologies, while also covering practical aspects such as server configuration, web hosting, cloud computing, and emerging domains like the Internet of Things (IoT) and edge computing. Emphasis is placed on the security of internet services, with a focus on HTTPS, SSL/TLS, and web application vulnerabilities. Through theoretical learning, hands-on labs, and real-world case studies, students will gain the critical skills needed to analyze, design, and implement secure and scalable internet-based systems in both enterprise and research environments.

2. General Objectives

- To provide the students with a comprehensive understanding of Internet architecture, components, and protocols—including TCP/IP, DNS, HTTP/HTTPS, and email protocols—for seamless global communication and system design.
- To equip the students with practical skills in configuring web servers, deploying internet services, managing domains, and developing secure web applications using modern technologies and frameworks.
- To acquaint the students with emerging trends such as cloud computing, CDNs, IoT, and edge computing, along with internet security practices essential for scalable and secure internet-based systems.

3. Methods of Instruction

Lecture, Discussion, Readings, Case Studies and Group Workshops

4. Contents in Detail

Specific Objectives	Contents
Explain about Internet Technology, including IP addressing and Routing Protocols.	Unit 1: Overview of Internet Technology (9 hrs) 1.1 Evolution of the Internet and the role of standard bodies (IETF, W3C) 1.2 Overview of Internet Architecture 1.3 Role of ISPs, NAPs, and IXPs 1.4 Tiered ISP architecture 1.5 TCP/IP, IP Addressing (IPv4 and IPv6) 1.6 Routing Protocols (Static, Dynamic Routing and BGP Basics) 1.7 Overview of Software-Defined Networking
Summarize various Internet Applications, Protocols and Services	Unit 2: Internet Applications, Protocols and Services (6 hrs) 2.1 Transport Layer and Protocols (TCP and UDP)

	2.2 Domain Name System (DNS) 2.3 Overview of Email Protocols (SMTP, POP, IMAP) 2.4 FTP, SFTP, and Cloud Storage Services 2.5 VoIP, FoIP and IP Interconnection 2.6 Web Services and APIs 2.7 CDN (Content Delivery Networks)
Apply the key concept of Web Technologies and Standards	Unit 3: Web Technologies and Standards (7 hrs) 3.1 Client-Server Architecture 3.2 HTTP2.0/HTTPS, QUIC, RESTful Services 3.3 Web Browsers and Rendering Engines 3.4 MIME Types and Content Negotiation 3.5 Modern Web Architectures (Microservices, Serverless)
Describe Web Server and Hosting Technologies	Unit 4: Web Server and Hosting Technologies (6 hrs) 4.1 Web Servers: Apache, Nginx, Microsoft IIS 4.2 Static vs. Dynamic Content 4.3 Domain Name Registration and Hosting 4.4 Web Server Mirroring 4.5 Web Server and Proxy Load Balancing
Describe internet security, secure communication protocols, web application security and standards	Unit 5: Internet Security (7 hrs) 5.1 Fundamental Cryptographic Concepts 5.2 Security Principles for Internet Services 5.3 Secure Communication Protocols (SSL/TLS) 5.4 Digital Certificate, HTTPS, Secure Email and DNSSEC 5.5 Web Application Security (OWASP Top 10) 5.6 Firewalls, IDS/IPS, Secure Tunneling (VPNs) 5.7 Standards and Frameworks (MITRE, NIST, ISO 27001, C2M2)
Explain Cloud and Edge Computing	Unit 6: Cloud and Edge Computing (6 hrs) 6.1 Introduction to Cloud Computing 6.2 Cloud Deployment Models: IaaS, PaaS, SaaS 6.3 Containerization and container-based services 6.4 Edge Computing and its Significance 6.5 Service Providers and Offerings (AWS, Azure, GCP)
Describe Internet of Things and Future Trends	Unit 7: Internet of Things (IoT) and Future Trends (4 hrs) 7.1 IoT Architecture and Communication Models 7.2 MQTT, ZeroMQ, CoAP, and IoT Security 7.3 Smart Cities, Wearables, and Connected Devices 7.4 Future Trends: Web 3.0, Blockchain, Quantum Internet

5. Case Studies

The Practical Sessions and Case Studies should cover the following works:

- Routing Protocols Deployments in Simulator (Packet Tracer/GNS)
- Setting up a Web Server in Linux and Windows
- Domain Name System Configuration
- Setting up Web/Proxy Server Load Balancer
- Setting up Docker container for Deploying Micro Services
- HTTPS and SSL Certificate Installation

- Case Study: Cloud Hosting Architecture for a Startup
- Case Study: DNS Attack and Mitigation Strategy

6. Evaluation system and Students' Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment and group workshop. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory			Semester-End examination	40
Attendance & Class Participation	10%	6		
Assignments	10%	6		
Presentations/Quizzes	10%	6		
Internal Assessment	50%	30		
Practical and Case Studies	20%	12		
Total Internal		60		
Full Marks: 60 + 40 = 100				

Student Responsibilities:

Each student must secure at least 60% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

References:

1. Kurose, J. F., & Ross, K. W. (2021). *Computer networking: A top-down approach* (8th ed.). Pearson.
2. Comer, D. E. (2018). *Internetworking with TCP/IP Volume One: Principles, protocols, and architecture* (6th ed.). Pearson Education.
3. Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer networks* (5th ed.). Pearson.
4. Forouzan, B. A. (2013). *Data communications and networking* (5th ed.). McGraw-Hill Education.
5. Buyya, R., Vecchiola, C., & Selvi, S. T. (2013). *Mastering cloud computing: Foundations and applications programming*. Morgan Kaufmann.
6. Newman, S. (2015). *Building microservices: Designing fine-grained systems*. O'Reilly Media.
7. Richardson, C., & Smith, F. (2016). *Microservice architecture: Aligning principles, practices, and culture*. O'Reilly Media.
8. RFCs (especially for HTTP, DNS, BGP, etc.)
9. W3C, IETF, OWASP documentation